A function can approach a value it never reaches

Nearness, not arrival, is the phenomenon we need to describe.

Watch the inputs move toward the target while the outputs settle near one value.

Two linked neighborhoods make approach measurable

A chosen output tolerance determines how close inputs must be.

- The output band is centered on the proposed limit.
- The input band is centered on the target.
- Every permitted input must land inside the output band.

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

For a line, the needed input band follows directly

For f(x)=2x+1 near x=3, choosing delta=epsilon/2 is sufficient.

- Start from the output error |(2x+1)-7|.
- Factor it as 2|x-3|.
- Require |x-3| < epsilon/2.

$$|f(x) - 7| = 2|x - 3| < \varepsilon$$

The limit records stable local behavior, not a point value

Controlled neighborhoods explain approach even when f(a) differs or is absent.

You can now test a proposed limit by connecting any requested output tolerance to a sufficient input tolerance.